

Pokhara University
Faculty of Management Studies

Course Code: MTHS213 174
Course Title: **Mathematics II**
Nature of the Course: Theory
Level: Bachelor

Credits: 3[Hrs]
Total Lectures: 48 hours
Program: BCSIT
Year: I Semester: II

1. Course Description

This course covers essential mathematical concepts starting with complex numbers, including their definition, algebra, properties, and graphical representation. It introduces De Moivre's theorem and applications. The course progresses to infinite sequences and series, focusing on convergence tests and specific tests like P-series and ratio tests. It explores antiderivatives, including definite and improper integrals, quadrature, rectification, and Beta and Gamma functions. Optimization for functions of several variables, partial differentiation, and finding maxima and minima are also covered. Students will study ordinary differential equations, including first-order solutions and second-order equations with constant coefficients, along with initial and boundary value problems. The course addresses number theory, including division, primes, GCD, LCM, and modular arithmetic, with applications in cryptology. Finally, it covers Fourier series and integrals, exploring even and odd functions, periodic functions, and Fourier coefficients, as well as sine and cosine integrals.

2. General objective

- Understand complex numbers, their properties, and algebraic operations.
- Proficiently analyze infinite sequences and series using convergence tests.
- Master antiderivatives, including definite and improper integrals, and apply Beta and Gamma functions in problem-solving.
- Solve ordinary differential equations and optimization problems using techniques like partial differentiation

3. Specific Objectives and Contents

Specific Objectives	Content
• Understand and define complex numbers, including integral powers of i. • Execute addition, subtraction, multiplication, and division of complex numbers. • Investigate properties of complex numbers, including conjugates and moduli. • Represent complex numbers using Argand's diagram, express in polar form, and apply De Moivre's theorem	Unit I: Complex Numbers [8 hrs.] 1.1 definition of a complex number, integral powers of i. 1.2 algebra of complex numbers (sum, difference, multiplication, division). 1.3 properties of complex numbers (without proof), conjugate of a complex number and its properties.



	1.4 modulus of a complex number and its properties (without proof), representation of a complex number by a point in a plane (Argand's diagram). 1.5 polar representation of a complex number. 1.6 square roots of a complex number (only Cartesian form) 1.7 De Moivre's theorem (statement only) and its application.
- Learn the concept and conditions of convergence tests for infinite series. - Use direct and limit comparison tests to determine series convergence or divergence. - Apply P-series, De Alembert's ratio, and alternating series tests to analyze series convergence.	Unit II: Infinite Sequence and Series [7 hrs.] 2.1. Introduction 2.2. Convergence test of infinite series (statement only). 2.3. Direct comparison test, Limit comparison test (statement only). 2.4. P-series test, De Alembert's ratio test, and Alternating series test (statement only).
- Learn the concepts, properties, and applications of definite and improper integrals. - Calculate areas (quadrature) and curve lengths (rectification) for functions $y=f(x)$ using antiderivative techniques - Study and apply the properties of Beta and Gamma functions in different mathematical contexts.	Unit III: Application of Antiderivative [7 Hrs.] 1.1. Definite integral 1.2. Properties of the definite integral 1.3. Improper Integral 1.4. Quadrature [$y = f(x)$] 1.5. Rectification [$y = f(x)$] 1.6. Beta and Gamma function.
- Learn the concept and importance of partial derivatives - Use rules of partial differentiation for accurate calculations - Identify maxima and minima for functions of two variables	Unit IV: Optimization: Functions of several variables [6 hrs] 4.1. Introduction 4.2. Partial derivative 4.3. Rules of partial differentiation 4.4. Maxima and minima for the function of two variables.
Understand the order and degree of differential equations	Unit V: Ordinary Differential Equation [7 Hrs.] 5.1. Introduction.



- Solve first-order and first-degree differential equations using various methods. - Solve second-order linear differential equations with constant coefficients and address initial and boundary value problem.	5.2. Order and degree of differential equation. 5.3. Solution of the first order and first-degree differential equation. 5.4. Variable separation, homogeneous, linear differential equation. 5.5. Second-order linear differential equation with constant coefficients. 5.6. Initial and boundary value problems.
- Identify prime numbers and apply them in problem-solving - Master division algorithm, GCD, and LCM for integer problem-solving. - Apply modular arithmetic in cryptography for secure message transmission	Unit VI: Integers and Division [6 Hrs.] 1.1. Introduction 1.2. Division, primes, the fundamental theorem of arithmetic (statement only) 1.3. The infinitude of primes 1.4. The division algorithm, GCD and LCM 1.5. Modular arithmetic 1.6. Application of congruence's Cryptology.
- Classify functions as even or odd and analyze their properties. - Represent periodic functions using Fourier series and determine coefficients. - Apply Fourier integrals to represent non-periodic functions	Unit VII: Fourier Series and Integrals [7 Hrs.] 1.1. Introduction 1.2. Even and odd function 1.3. Periodic function 1.4. Fourier series and Fourier coefficients (without proof) 1.5. Fourier sine and cosine series 1.6. Fourier integral 1.7. Fourier sine and cosine integral

4. Methods of Instruction

- Lecture
- Group discussion
- Question-answers
- Demonstration and discussion
- Presentations
- Guest lectures



- Group work/project work
- Problem solving
- Simulation
- Tutorials

5. Evaluation System and Student's Responsibilities

Evaluation System

In addition to the formal exam(s), the internal evaluation of a student may consist of quizzes, assignments, class participation, etc. The tabular presentation of the internal evaluation is as follows.

External Evaluation	Marks	Internal Evaluation	Weight	Marks
Semester-End examination	50	Theory		50
		Attendance & Class Participation	10%	
		Assignments	20%	
		Presentations/Quizzes	10%	
		Internal Assessment	60%	
Total External	50	Total Internal		50
Full Marks: 50 + 50 = 100				

Student's Requirement

Each student must secure at least 45% marks separately in both internal assessment and practical evaluation with 80% attendance in the class to appear in the semester-end examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear for the Semester-End Examinations. Students are advised to attend all the classes, formal exams, tests, etc., and complete all the assignments within the specified period. ***Students are required to complete all the requirements defined for the completion of the course.***

6. Prescribed Books and References

Text Books

1. Kreyszig, E. (2020). *Advanced Engineering Mathematics*. New Delhi: John Wiley & Sons Inc.
2. Thomas, G. B. Jr., & Finney, R. L. (2003). *Calculus and Analytical Geometry*. New Delhi: Narosa Publishing House.
3. Rosen, K. H. (2003). *Discrete Mathematics and its Applications* (5th ed.). McGraw Hill Companies.

Reference

1. Shrestha, K. K., & Thagurathi, R. K. (Year of publication). *Applied Mathematics*. Kathmandu, Nepal: Buddha Publication

